

Profiling data

FIELD NAME	TOTAL	EMPTY/MISSING	ACTUAL	COMPLETENESS	CARDINALITY	UNIQUENESS	DISTINCTNESS	MIN	MAX	MODE	AVG	MEDIAN	VALIDITY REGEX ?	VALIDITY (%)	
OrderID	25	1	24	96.0%	22	88.0%	91.7%	101	124	101.0	112.21	112.50	Regex ...	100.0%	
CustomerName	25	2	23	92.0%	21	84.0%	91.3%	Alice Brown	peter@exampl...	John Doe	-	-	[\s a-zA-Z]	95.7%	>
Product	25	2	23	92.0%	21	84.0%	91.3%	Bed	Webcam	Laptop	-	-	Regex ...	100.0%	
Category	25	1	24	96.0%	8	32.0%	33.3%	Accessories	Software	Electronics	-	-	Regex ...	100.0%	
Price	25	2	23	92.0%	18	72.0%	78.3%	-50	50000	20	2667.62	150	Regex ...	100.0%	
Quantity	25	1	24	96.0%	8	32.0%	33.3%	-1	1000000	1	43479.89	1	Regex ...	100.0%	
OrderDate	25	1	24	96.0%	22	88.0%	91.7%	01/18/2023	27-01-2023	2023-01-15	-	-	Regex ...	100.0%	
PaymentStatus	25	1	24	96.0%	7	28.0%	29.2%	Error	Unpaid	Paid	-	-	Regex ...	100.0%	
ShippingCity	25	1	24	96.0%	23	92.0%	95.8%	Austin	new york	Fort Worth	-	-	Regex ...	100.0%	

Defect Analysis

FIELD NAME	DATA TYPE	OUTLIERS (IQR)	TOP 5 PATTERNS
OrderID	Decimal	-	sss.s100.0%
CustomerName	String	-	ssss sssss21.7% sss sssss13.0% sssss ssss13.0% ssss sss8.7% sssss sssss8.7%
Product	String	-	ssssss21.7% ssssss17.4% sssss13.0% ssss13.0% ssssssssss8.7%
Category	String	-	ssssssssss45.8% ssssssssss33.3% ssssss8.3% ssssssss8.3% ssssssssss4.2%
Price	Integer	120050000	sss39.1% ss30.4% sssss13.0% \$sss4.3% sssss4.3%
Quantity	Integer	-14101000000	s79.2% sss4.2% -s4.2% ss4.2% s.s4.2%
OrderDate	Date	-	ssss-ss-ss91.7% ss/ss/ssss4.2% ss-ss-ssss4.2%
PaymentStatus	String	-	ssss70.8% sssssss8.3% sssss8.3% ssssssss4.2% ssssss4.2%
ShippingCity	String	-	ssssss20.8% ssssss16.7% sss ssss12.5% sss sssssss8.3% ssss sssss8.3%



Q&A

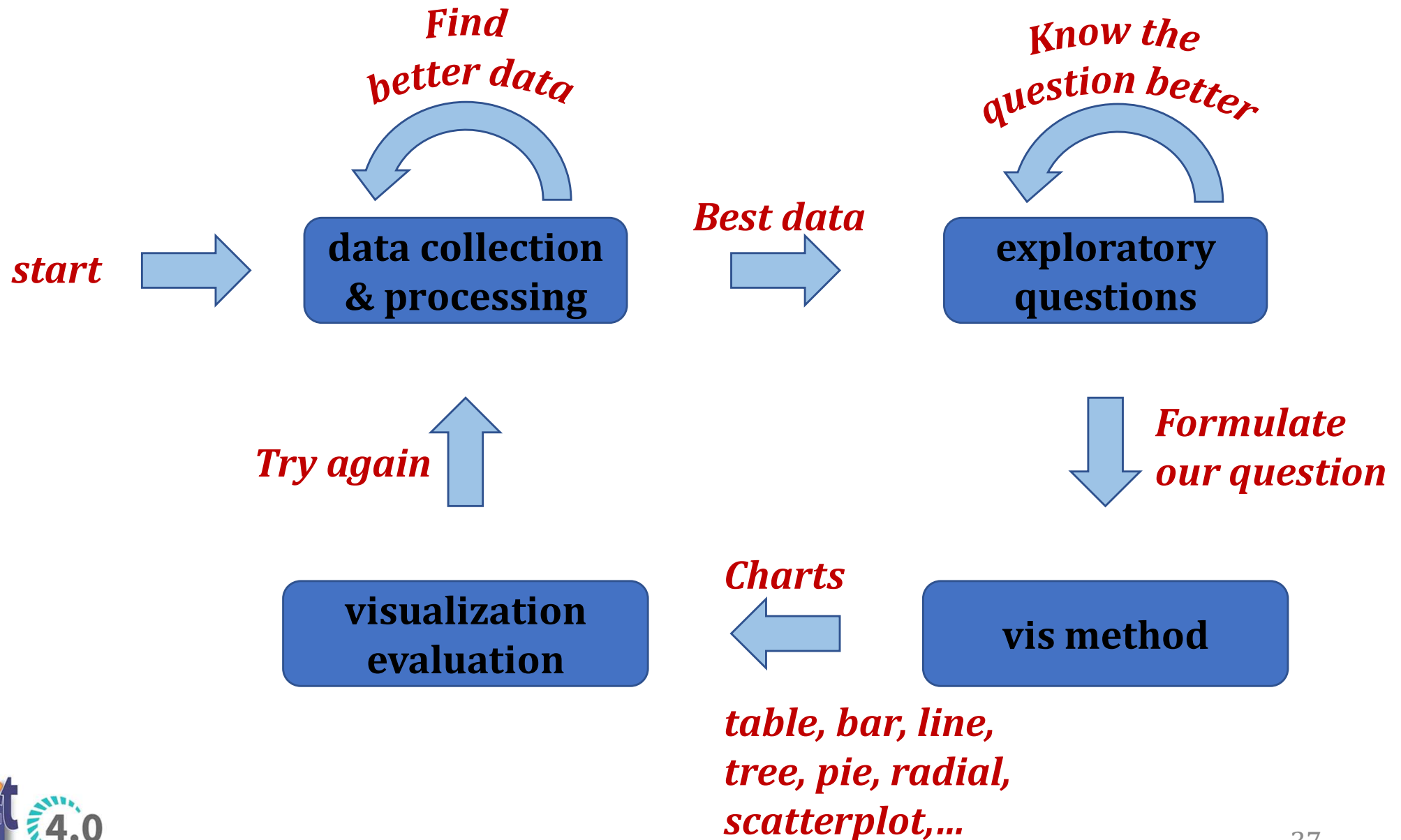


Chapter 4

Task Abstraction



Visualization workflow



Know Your Ingredients



- Before asking a question, audit your dataset for two types of columns:
 - **Measures (What - Q)**
 - Quantitative values you can calculate
 - Examples: Sales, Profit, Temperature, Heart Rate, Debt Amount.
 - **Dimensions (Context – C, O)**
 - Qualitative categories used to slice data.
 - Examples: Date, Region, Product Category, Risk Level, Gender.

*A "Good Question" always combines **1 Measure** with **multiple Dimensions**.*

Methods



- Actionable User Story Formula
- 5 Whys Drill-Down
- Inverted Pyramid
- 3-Tier Logic Chain (What → Why → Next)

User Story Formula

- *Methods*

As a *[Role]*, I need *[Metric + Breakdown]* so I can *[Decision/Action]*

- Example

- As *Marketing Manager*, I need *ROAS (Return On Ad Spend)* by *campaign & device* so I can *shift budget to campaigns > 4.0 ROAS*
- As *Store Manager*, I need *top 10 slow-moving SKUs* by *store* so I can *apply 40% discount or return to supplier*
- As *Credit Risk Officer*, I need *overdue loans > 90 days* by *province & customer segment* so I can *prioritize collection teams in high-risk areas*
- As *HR Lead*, I need *employee satisfaction score* by *department & tenure* so I can *design retention program for high-risk groups (<1 year)*

5 Whys Drill-Down

- **Methods**

*Keep asking “**Why?**” or “**What drives this?**” until actionable insight*

- Example

- Vaccine logistics

- Surface: Wastage 14%
 - Why: Mostly rural clinics
 - Why: Delivery > 48 h
 - Why: No temperature alert system

→ **Deep question:** “How does delivery time vs temperature deviation affect wastage by vaccine type & province?”

- Cafe chain

- Surface: Weekend sales flat
 - Why: Saturday lunch hour weak
 - Why: Long waiting time (>12 min)
 - Why: Only 2 baristas on shift

→ **Deep question:** “Staffing level vs order volume vs wait time correlation by hour & day”

Inverted Pyramid



- ***Methods***

- *Top (5–10 sec glance): KPIs*
 - Current revenue, % vs target, MoM change, Top 3 alerts
- *Middle (context & trends):*
 - Revenue trend last 12 months
 - Category / region contribution (bar / heatmap)
 - Top vs bottom performers comparison
- *Bottom (details on demand):*
 - Drill-down table: product × store × day
 - Customer list with filters

3-Tier Logic Chain



- *Methods*

- *Tier 1: Descriptive* (Status → **WHAT**)
- *Tier 2: Diagnostic* (Reason → **WHY**)
- *Tier 3: Prescriptive* (Action → **NEXT**)

- Example

- Hospital:
 - **What**: Patient wait time > 45 min (target 30 min)
 - **Why**: Peak hours 8–10 AM, mainly blood test queue
 - **Next**: Add 2 temporary nurses 8–11 AM starting next week
- School:
 - **What**: Dropout rate Q4 = 8.2%
 - **Why**: Highest in online Math courses for grade 10
 - **Next**: Record short explainer videos + assign peer tutors

Assignment #4.1

Formulating Powerful Questions for Data Visualization

- **Business Scenario**

- You are a **Data Analyst** at a small e-commerce store selling:
 - Electronics & accessories
 - Furniture & home items
 - Some office & miscellaneous products
- **Current company priorities (2023):**
 - Achieve stable and sustainable revenue growth
 - Reduce incomplete / failed orders (Pending, Unpaid, Refunded)
 - Better understand customer purchasing behavior by
 - Product category
 - Shipping city
 - Time (month, day of week)

Assignment #4.1

Formulating Powerful Questions for Data Visualization

- **Create these calculated fields if needed**
 - TotalRevenue = Price × Quantity
 - Month / YearMonth
 - DayOfWeek
 - OrderSuccess (1 = Paid, 0 = otherwise)
- **Task**
 - Task 1: Write at least 2 high-quality User Stories
 - Task 2: Apply 5 Whys to 2 Business Pain Points
 - Task 3: Select & List Your Top 10 Questions
 - Task 4: Build 3-Tier Logic Chains for 2 Key KPIs

Task abstraction



- ***Task abstraction*** – rather than thinking of *domain-specific* tasks, think of *abstract* tasks
 - Ex:
 - *Domain-specific*: contrast the prognosis of patients who were intubated in the ICU for more than one month to patients hospitalized within the first week
 - *Abstraction*: compare values between two groups

Task abstraction



- Transforming task descriptions from *domain-specific* into *abstract* form allows us to reason about *similarities* and *differences* between them
 - Ex:
 - *Domain-specific*: contrast the prognosis of patients who were intubated in the ICU for more than one month to patients hospitalized within the first week
 - *Domain-specific*: comparison of retail sales for member and non-member customers in January
 - => *Abstraction*: compare values between two groups

Task abstraction



• Task = **action** + **target**

▪ Action verbs and target nouns

▪ **Action** verbs

- Analyze
 - Consume
 - Produce
- Search
 - Lookup, browse, locate, explore
- Query
 - Identify, compare, summarize

▪ **Target** nouns

- All data
 - Trends, outliers, features
- Attributes
 - One
 - Distribution, extremes
 - Many
 - Dependency, correlation, similarity
- Network data
 - Topology: paths
- Spatial data
 - Shape

Three levels of actions

•*Analyze* - high-level depiction of the data

- Examining and evaluating information or data to *gain insights, draw conclusions, or make decisions*
- Ex: A manager needs to **analyze sales data** to identify *trends* and *patterns*: => **Consume > Discover**
 - Sales growth rates of top-performing products
 - Determining the impact of marketing campaigns

•*Search* – looking for specific-information (*items, attributes, links,...*)

- Looking for specific questions or requests to retrieve *summaries* and *abstractions*
- Ex: a manager must *compare the sales* of a products with similar other products => **Browse**

•*Query* – looking for characteristics of items and nodes

- Posing specific questions or requests to retrieve summaries and abstractions.
- Ex: A manager needs a report on: => **Summarize**
 - The average sale of a product
 - Expenses within a specific period
 - Revenue of a specific months

Actions - **Analyze**

• **Analyze** – *consume* information already generated or *produce* new material for some purpose

- **Consume**

- **Discover** – visualization to find new knowledge that was not previously known
 - Relationship, Patterns, Outliers, ...
 - Required charts that facilitate exploration and reveal hidden information
 - Ex: Scatter plots, node-link (network) diagram, stacked bar, and pie chart
- **Present** – help user communicate something known to someone
 - Explain and discuss.
 - Requires clear and easily understandable visualizations
 - Ex: Bar charts, line charts, pie charts, and infographics
- **Enjoy** – casual interaction within a visualization
 - Social, study, etc.
 - Ex: infographics and storytelling

Actions - **Analyze**

- **Analyze** – *consume* information already generated or *produce* new material for some purpose
 - **Produce**
 - *Annotate* – add graphical or textual annotations to pre-existing visualization
 - *Record* – persist visualization elements
 - *Derive* – produce new data elements based on existing data elements (*transformation*)

Actions – Search

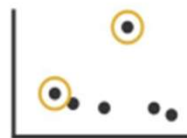
- Each analysis goal requires the user to search for items of interest within the visualization
 - **Lookup** – involves finding specific information or a known item
 - Ex: You want to check the price of the house at *'123 A Street'*
 - **Locate** – finding the physical or digital position of information or an item
 - Ex: You want to find all houses *priced below 2 billion VND*.
 - **Browse** – finding information or an item with specific characteristics
 - Ex: You want to buy a house *in District 1*
 - **Explore** – investigating and discovering information
 - Ex: You are new to the city and want to find a *'livable'* area

	Target known	Target unknown
Location known	 <i>Lookup</i>	 <i>Browse</i>
Location unknown	 <i>Locate</i>	 <i>Explore</i>

Actions – Query

- Finding specific information or subsets of data from datasets based on predefined criteria or conditions
 - **Identify** – extract characteristics or references from *a single target (one)*
 - Ex: After finding a house you like, ‘*How many bedrooms*’ does this house have?’
 - **Compare** – compare *multiple targets* to each other (*some*)
 - Ex: Select three houses you like and *compare their prices* to see which one is cheaper
 - **Summarize** – provide a comprehensive view of *all the data* (*overview*)
 - Ex: The *average price* of the entire real estate market in this city.

→ Identify



→ Compare



→ Summarize



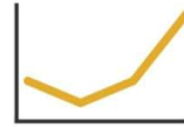
Targets

- **Targets** are the aspects of the data of interest to user

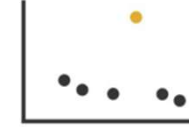
- Target that apply to *all data*

- Trends (*patterns*)
- Outliers
- Features

→ Trends



→ Outliers



→ Features



- If data carries *attributes*

- Single attribute (*distribution, extremes*)
- Many (*dependencies, correlation, similarity*)

→ One

→ Many

→ Distribution



→ Dependency



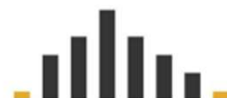
→ Correlation



→ Similarity



→ Extremes



User visualization

Actions

→ Analyze

→ Consume

→ Discover



→ Present



→ Enjoy



→ Produce

→ Annotate



→ Record



→ Derive



→ Search

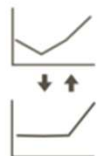
	Target known	Target unknown
Location known	••• Lookup	••• Browse
Location unknown	<•••> Locate	<•••> Explore

→ Query

→ Identify



→ Compare



→ Summarize



Targets

→ All Data

→ Trends



→ Outliers



→ Features



→ Attributes

→ One

→ Distribution



→ Extremes

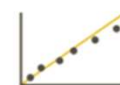


→ Many

→ Dependency



→ Correlation



→ Similarity



→ Network Data

→ Topology



→ Paths



→ Spatial Data

→ Shape



What?

Why?

How?

Domain-Task



- **Ngữ cảnh** - *quản lý bán hàng* của cửa hàng thương mại điện tử nói:

"Tôi muốn biết tại sao doanh thu tháng này giảm so với tháng trước, đặc biệt là ở một số nhóm sản phẩm và một số thành phố giao hàng. Tôi cần hiểu rõ để quyết định có nên tăng khuyến mãi cho nhóm sản phẩm nào hoặc điều chỉnh vận chuyển ở thành phố nào không."

⇒ **Domain-specific task**

"Tìm hiểu lý do doanh thu giảm so với tháng trước, tập trung vào sản phẩm và khu vực giao hàng, để ra quyết định hành động kinh doanh."

Bước 1: Phân tích câu hỏi domain

Domain-task: *"Tìm hiểu lý do doanh thu giảm so với tháng trước, tập trung vào sản phẩm và khu vực giao hàng, để ra quyết định hành động kinh doanh."*

Loại chi tiết domain:

- Xem sự thay đổi theo thời gian.
- So sánh giữa các nhóm (sản phẩm, khu vực).
- Tìm nguyên nhân.
- Hỗ trợ ra quyết định.

Bước 2: Xác định Action và Target

Cấp độ	Phân loại	Lý do	Action + Target
Analyze	Consume → Discover	Người dùng muốn phát hiện insight mới (<i>tại sao giảm, nhóm nào gây ra</i>) để ra quyết định kinh doanh.	Discover + Trends + Dependence
Search	Explore → Browse → Locate	- Không biết chính xác nhóm sản phẩm/thành phố nào gây giảm (Explore) - Tập trung vào các category/region có vấn đề (Browse) - Tìm vị trí cụ thể của outlier hoặc nhóm thấp (Locate)	Explore → Browse → Locate + Features / Outliers
Query	Compare → Summarize → Identify	- So sánh doanh thu giữa các tháng, giữa các category, giữa các thành phố. - Tổng hợp mức giảm trung bình. - Xác định chi tiết các nhóm cụ thể gây vấn đề.	Compare + Summarize + Identify + Trends / Distribution / Dependence

Bước 3: Task Abstraction

- **Discover + Trends + Dependence**

- Mục tiêu cao nhất: Phát hiện xu hướng thay đổi doanh thu và mối phụ thuộc giữa doanh thu với category/region theo thời gian

- **Explore + Browse + Features**

- Cách tìm: Khám phá tự do toàn bộ dữ liệu, duyệt qua các nhóm category và region để tìm đặc trưng nổi bật

- **Locate + Compare + Outliers**

- Sau khi duyệt: Tìm vị trí các nhóm/category/region bất thường, so sánh mức độ giảm giữa chúng

- **Summarize + Compare + Distribution**

- Truy vấn: Tổng hợp và so sánh phân bố doanh thu theo category/region/thời gian

→ **Full - task abstraction**

Discover + Explore + ~~Browse~~ + ~~Locate~~ + Compare + Summarize + Trends / Dependence / Distribution / ~~Outliers~~

Bước 3: Task Abstraction

- **Discover + Trends + Dependence**

- Mục tiêu cao nhất: Phát hiện xu hướng thay đổi doanh thu và mối phụ thuộc giữa doanh thu với category/region theo thời gian

- **Explore + Browse + Features**

- Cách tìm: Khám phá tự do toàn bộ dữ liệu, duyệt qua các nhóm category và region để tìm đặc trưng nổi bật

- **Locate + Compare + Outliers**

- Sau khi duyệt: Tìm vị trí các nhóm/category/region bất thường, so sánh mức độ giảm giữa chúng

- **Summarize + Compare + Distribution**

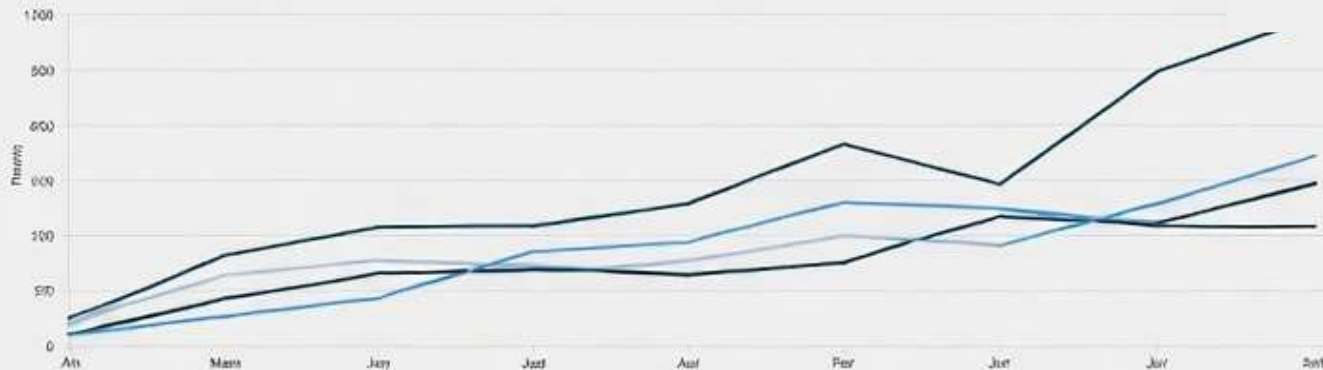
- Truy vấn: Tổng hợp và so sánh phân bố doanh thu theo category/region/thời gian

→ **General task abstraction**

Discover + Explore + Compare + Trends / Dependence / Distribution

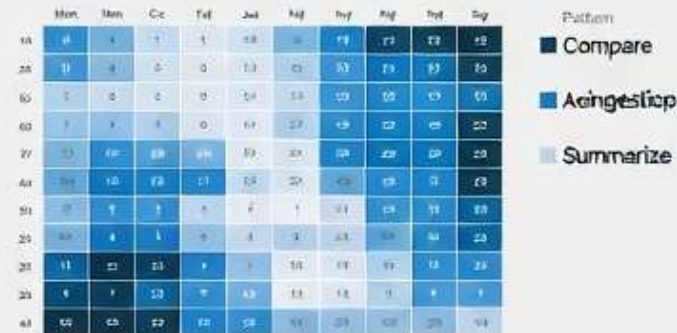
Bước 4: Chọn biểu đồ

Line chart



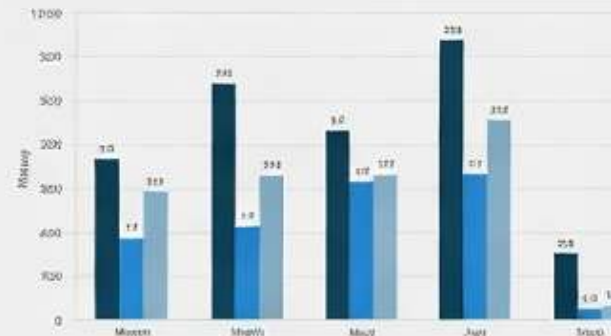
**Trend
Compare**

Heatmap



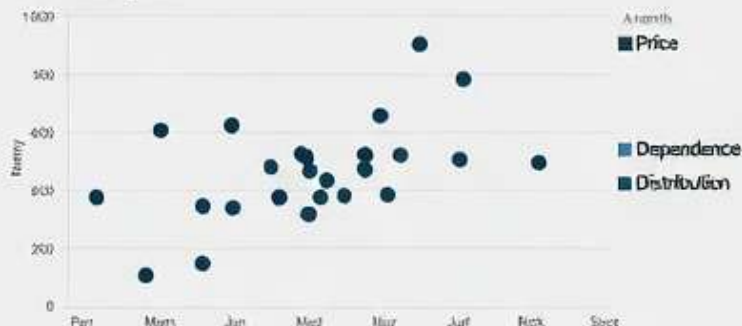
**Dependence
Summarize
Outliers
Distribution**

Bar chart



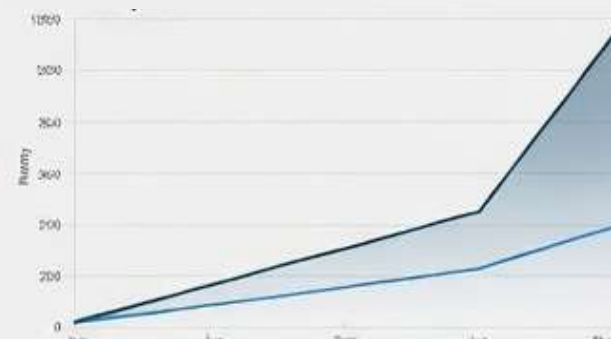
**Compare
Outliers
Identify**

Scatter plot

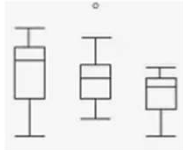
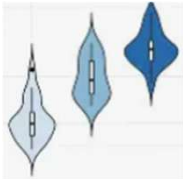
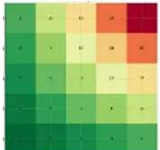


**Outliers
Distribution
Compare**

Area chart



**Trend
Compare**

Abstraction task	Chart type	Chart
Top/bottom	Bar chart Pie Tree map	  
Trend	Line chart Area chart	 
Distribution	Histogram Box plot Violin plot Heatmap	   
Dependency/ Correlation/ Similarity	Scatter plot Bubble chart Tree map	  
Outlier	Scatter plot Heatmap	
Features	Line chart	

How to Design Vis. Idioms?

Idiom	Chart name
What	Data abstraction
How	Encoding
	Manipulate
	Facet
	Reduce
Why	Task abstraction
Scale	Amount of key and level

How to Design Vis. Idioms?



- There are in general four choices:
 - Encode
 - Manipulate
 - Facet
 - Reduce

How to Design Vis. Idioms?

• *Encode*: The way of encoding data within a view has five choices for how to arrange data spatially:

- Express *values*
- Separate, order, and align *regions*
- use given *spatial data*.

→ Express



→ Separate



→ Order



→ Align



→ Use



How to Design Vis. Idioms?

- *Encode*: Also includes how to map data (from *categorical* and *ordered* attributes) with all the non-spatial visual channels (e.g., color, size, angle, shape, etc.)

→ Color

→ Hue



→ Saturation



→ Luminance



→ Shape



→ Size, Angle, Curvature, ...



→ Motion

Direction, Rate, Frequency, ...



How to Design Vis. Idioms?

- *Manipulate*: This family has the choices of *changing* any aspect of the view, *selecting* elements from within a view, and *navigating* to change the viewpoint within the view.

➔ Change



➔ Select



➔ Navigate



How to Design Vis. Idioms?

- *Facet*: The family of how to facet data between views has choices for how to *juxtapose* and coordinate multiple views, how to *partition* data between views, and how to *superimpose* layers on top of each other.

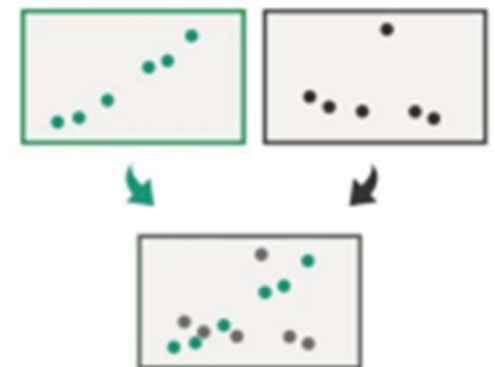
➔ Juxtapose



➔ Partition



➔ Superimpose



How to Design Vis. Idioms?

- *Reduce*: The family of how to reduce the data shown has the options of *filter* data away, *aggregate* many data elements together, and *embed* focus and context information together within a single view.

➔ Filter



➔ Aggregate

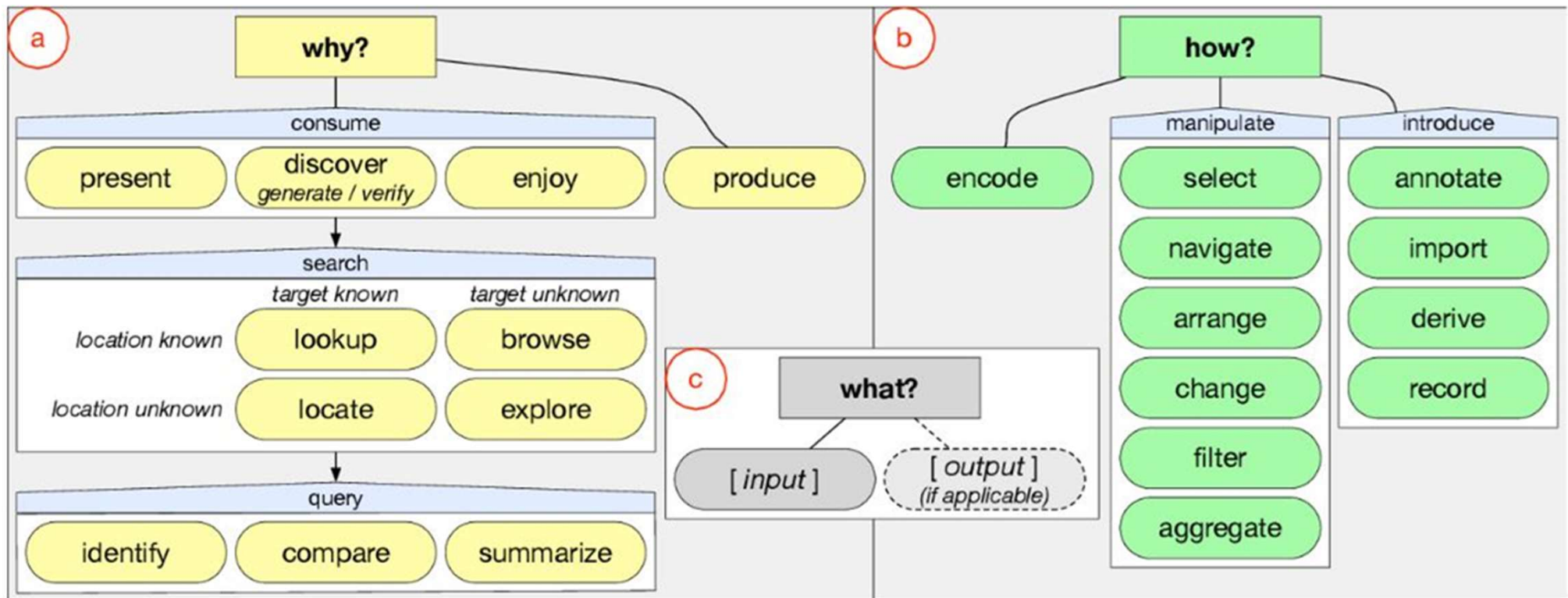


➔ Embed



How to Design Vis. Idioms?

- Three levels of actions that define *user goals*



Bài tập Task-abstraction



Domain-task: (Dataset: *retail_sales_dataset.xls*)

- So sánh doanh số bán của tất cả sản phẩm giữa 2 nhóm khách hàng Nam/Nữ
- Phân tích xu hướng mua hàng của khách hàng theo từng nhóm tuổi
 - Dưới 18 tuổi
 - Từ 18 – dưới 40 tuổi
 - Trên 40 tuổi

• **Yêu cầu:**

- Data abstraction cho dữ liệu liên quan đến domain-task
- Task abstraction cho 2 domain-task trên



Q&A

